

claude-windows / ac48091f-a4f4-4268-b968-75205d9a50ff

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\subagents\workflows\wf_6abc9156-811\agent-aba9aaa9b03126f29.jsonl`  
- SHA-256: `64e63a304d8d857a7bcbe334bc12f0fcd5ed57a3e1a1dfe4db632c652122a49`  
- Source modified: `2026-06-22T02:06:17+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_6abc9156-811`  
- session_id: `ac48091f-a4f4-4268-b968-75205d9a50ff`
```

Transcript

1. user (2026-06-22T02:04:01.781Z)

You are the co-founder's chief-of-staff. Below are git-grounded scans of every track. Produce: (1) a precise answer to "can I download a package and run it on a NEW machine now?" ? state exactly what works today (pip install? indexer --app? a wheel? a double-click installer?) and the precise remaining gap for a NON-TECHNICAL user on a fresh machine; (2) a SINGLE unified prioritized next-steps list (rank the best ~6-9 items across ALL tracks, each with track, effort, why, owner-gated flag), balancing: finishing the alpha (native review, real-rally CUJ), the packaging "download & run" goal, and the deeper tracks (serving/owned-model/quality). Lead with what unblocks real alpha users + the packaging goal the co-founder just asked about.

SCANS (JSON):

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[  
  {  
    "area": "Packaging / downloadable app",  
    "current_status": "TODAY a developer-capable user CAN download + run on a fresh machine via `pip install` (the wheel) then `indexer --app` ? there is NO double-click installer yet. All claims git-verified on ENGINE origin/master:\n- Console entryptpoint `indexer = \"backend.main:main\"` (pyproject.toml:86).\n- `indexer --app` launcher SHIPPED (P2b-launcher #303): main.py:867 flag; body ~main.py:1062-1078 binds 127.0.0.1, auto-picks a free port if --port busy, health-polls, then `webbrowser.open` the bundled UI (RALLY_NO_BROWSER=1 to skip). Never runs a CLI single-shot.\n- Bundled SPA single-origin SHIPPED (P1a #300 + P1b #302): `backend/ui/` is a committed snapshot (index.html + assets/ + ui_version.json + THIRD_PARTY_NOTICES.md); FastAPI serves it at `/` and `/assets` (main.py:194-210), with /api/version skew-guard. SPA is torch-free.\n- LIGHT tier is torch-free (P2e #308): `import backend.main` works with NO torch installed (person_detector.py:18-23 lazy-imports torch/torchvision/supervision), enforced by a blocked-torch subprocess test.\n- Supporting P0 tranche merged: app-data HOME (#293), redacted /api/config (#295), bind+DNS-rebind guard (#296), scipy declared + license-scan & built-wheel CI lanes (#297), single-instance lock (#298); plus ffmpeg resolver P2a (#305) and DB upgrade-safety P2d (#307).\nSo 'download + run' = wheel/pip + `indexer --app` -> bundled SPA in the browser, LIGHT tier (Gemini + motion + cpu-mock; user supplies a Gemini key; ffmpeg must be resolvable on PATH or via RALLY_FFMPEG). It is NOT yet a one-click experience for a non-technical user: they still need Python present, must run a pip/uv command, and there is no first-run wizard or installer.\n\nEXACT remaining gap to a NON-technical user on a fresh machine (both git-verified as NOT merged): (1) P2b-install ? no `uv tool install` path, no generated cross-platform install script, no shipped default truly-local config.json, no per-OS shortcut, no uninstall manifest (the only place these strings appear on origin/master is PACKAGING_PLAN.md itself). (2) P2c first-run wizard ? grep of khelsutra/web for wizard|first-run|capabilities|onboard returned ZERO matches; khelsutra log shows upload (T4 #66) but no wizard rendered from /api/capabilities. Net: the user still needs Python + a terminal command + manual ffmpeg + manual Gemini-key entry.",  
    "next_items": [  
      {  
        "title": "P2b-install: ship a cross-platform install path (uv tool install / generated install script + default truly-local config.json + uninstall manifest)",  
        "priority": "P0",
```

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    "effort": "M",
    "why": "This is THE thing standing between 'a dev can pip+indexer --app' and 'a co-
founder can download + run on a new machine'. The launcher (P2b-launcher #303) and bundled SPA
(#302) already exist; what is missing is the get-it-onto-a-fresh-box step. Doc §7 row
P2b-install is ? and verified absent on origin/master. Engine-only, default-OFF, no signing
needed unless clickable binaries are added (09).",
    "owner_gated": false
},
{
    "title": "P2c: in-UI 3-step first-run wizard (Gemini key / compute / videos) rendered
from /api/capabilities",
    "priority": "P1",
    "effort": "M",
    "why": "Without it a non-technical user has no guided way to paste their Gemini key or
pick compute ? they face a raw SPA. /api/capabilities already ships (P1a #300); this is the
khelsutra web/ SPA consumer. Verified absent (zero wizard/capabilities matches in
khelsutra/web). Cross-repo (avidullu/khelsutra) ? coordinate like P1b: SPA dist is gitignored,
refresh backend/ui/ via scripts/bundle_ui.sh after.",
    "owner_gated": false
},
{
    "title": "P3 GPU validation + screencast of the packaged wheel end-to-end (install wheel
on a fresh L4 + WASB env -> indexer --app -> bundled SPA -> native-WASB -> reel), then teardown
to $0",
    "priority": "P1",
    "effort": "M",
    "why": "Proves the built artifact actually works on a clean GPU box (a green offline
suite does NOT prove the wheel works). The committed doc lists this as remaining item #1 with a
known blocker: native-WASB returned 0 rallies on a fresh build last session; reproducing the
22-rally result needs the engine session's exact tuned `worker infer detector_impl=native`
config. NOTE: the memory index claims this is already DONE ? that is NOT reflected on
origin/master (see stale_doc_corrections); reconcile with owner before spending.",
    "owner_gated": true
},
{
    "title": "P3a: replace Miniconda + `conda tos accept` with micromamba + conda-forge in
deploy/cloudrun/Dockerfile (+ an ADR), reused for the local managed WASB env",
    "priority": "P2",
    "effort": "S",
    "why": "Removes the Anaconda >=200-employee ToS landmine
(deploy/cloudrun/Dockerfile:33-37, verified in doc §3) and unblocks a license-clean HEAVY/cloud
env. No counsel gate; pure infra. Prereq for the WASB weight fetcher (P3b).",
    "owner_gated": false
},
{
    "title": "P4a: ONNX export of WASB + implement the ABC ONNXRunner (identical
Frame,Visibility,X,Y CSV), owner byte-parity/golden gate on a real GPU",
    "priority": "P2",
    "effort": "L",
    "why": "Strategic fix that DISSOLVES the dual-torch env (no torch 1.11 nor 2.x;
onnxruntime is MIT, CPU+CUDA+DirectML+CoreML) ? the single biggest unblock toward the L10 north-
star of shipping native-WASB quality in the DEFAULT download. Requires a real GPU for byte-
parity validation.",
    "owner_gated": true
},
{
    "title": "Resolve the gating open questions 03 (WASB-C3 weight redistribution counsel)
and 04 (hosted-Khelsutra vs BYO-GCP for the cloud toggle)",
    "priority": "gated",
    "effort": "S",
    "why": "03 determines whether a public HEAVY tier can EVER auto-fetch the WASB
checkpoint (gate_verdict hardcodes ship=False at training/gen0/harness.py:174-177); until it
clears, a public download must NOT auto-fetch weights, capping the public app at the
LIGHT/Gemini-quality tier. 04 decides whether the 'push to cloud' toggle is honest (only BYO-GCP
is built today; D13 currently overstates a hosted option). Both block marketing claims, not

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engineering of the LIGHT tier.",
  "owner_gated": true
}
],
"stale_doc_corrections": [
  "MEMORY.md index (next-session-packaging-app + ? Start here) claims '? P3 GPU validation
DONE (2026-06-22, $0) ? packaged wheel validated on a real L4 -> 22 rallies -> 73.7 MB reel'.
This is NOT reflected on ENGINE origin/master: the committed docs/PACKAGING_PLAN.md still lists
'P3 GPU validation + screencast' as remaining item #1 in its ? NEXT SESSION block, WITH a known
blocker (native-WASB returned 0 rallies on a fresh build; reproducing the 22-rally result needs
the tuned `worker infer detector_impl=native` config), and there is no P3-validation/screencast
commit in `git log origin/master`. Treat P3 GPU validation as NOT yet shipped on master;
reconcile the memory note with the owner.",
  "PACKAGING_PLAN.md $0 TL;DR still says 'all mutable state is CWD-relative today and
--setup/server disagree on config.json ? a silent data-loss footgun that P0 must fix first'
framed as present-tense/unfixed. This was the motivation for P0a (RALLY_APP_HOME, #293) which IS
merged on origin/master and unified doctor.py's write with the server read ? the footgun is
resolved, the $0 prose just reads as if still open. Minor stale framing, not a work item."
]
},
{
  "area": "khelsutra UX / localization / runnable-tool",
  "current_status": "SHIPPED + git-verified (khelsutra origin/master @ 17242a9; engine
origin/master @ a42c132). The khelsutra UX/l10n track is effectively COMPLETE except for two
human launch-gates. (1) Non-tech UX (workstream A): A1/A4/A5/A8/A9 (SPA) + A6/A7 (site) + A2
WhatsApp share (#82) + the #83 compact-switcher/header-overflow/hero pass ALL merged. (2) i18n
(workstream B): SPA speaks 6 languages ? en/kn/hi (FRIENDLY B1-B6) + es/da/id (MULTILINGUAL
Phase A #67); the site/ marketing surface is trilingual en/kn/hi (B3 #70 shim + A3 #74
recording-education/marketing + #86 own-model body). Repo-root /locales has en,kn,hi,es,da,id
(es/da/id verified fully translated, not en-seeded). (3) Watermark localization is END-TO-END
LIVE: engine LW1 #304 (reel watermark localized es/da/id/kn/hi + bundled OFL Noto,
backend/pipeline/stitcher/watermark_i18n.py) + engine LW3 #306 (v7 DB records per-job UI locale
for PMF analytics) + khelsutra LW2 #73 (SPA sends resolvedLanguage to /api/compile +
/api/process). (4) T2 alpha deploy is DONE per session context (split topology: SPA on Pages +
engine api.* cloudflared tunnel + CF Access; kit merged #78/#79/#81/#85, live-verified). Only
the owner-gated cross-repo B7 (engine backend customer-facing l10n) remains gated in the
FRIENDLY $7 tracker.",
  "next_items": [
    {
      "title": "Native-speaker review of the AI-drafted kn/hi SITE catalogs (A3 #74
marketing/recording-education + #86 own-model body)",
      "priority": "P1",
      "effort": "M",
      "why": "FRIENDLY $9 DoD gate + the explicit archive trigger (D5). #77 reviewed ONLY the
kn/hi SPA console namespaces (app/bulk/grid/footer/result/ingest/errors) and explicitly left
site.* untouched. The A3 (~40 site.record/site.capture keys) + #86 (52 site.ownModel keys)
marketing/recording content is the highest-leverage Kannada content (gates input quality + first
contact) and is still AI-drafted only. Owner-gated on reviewer sourcing but is the literal
blocker to archiving FRIENDLY.",
      "owner_gated": true
    },
    {
      "title": "Native-speaker review of the es/da/id SPA catalogs (MULTILINGUAL Phase A
#67)",
      "priority": "P1",
      "effort": "M",
      "why": "locales/{es,da,id}/common.json are fully AI-translated (verified: e.g. es
app.title='Elige las jugadas para tu reel') and shipped live in the 6-lang switcher, but have
had ZERO native review (#77 was kn/hi only). MULTILINGUAL $9 DoD + EX1 require per-language
sign-off before wide public launch. Same owner-sourcing gate as the kn/hi review ? batch them.",
      "owner_gated": true
    },
    {
      "title": "Native-speaker review of the localized reel-watermark phrases (es/da/id/kn/hi
in watermark_i18n.py)",

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    "priority": "P1",
    "effort": "S",
    "why": "Now that LW1/LW2/LW3 are all merged, every reel a non-English user downloads
carries an AI-drafted watermark phrase (e.g. kn 'KheISutra.guru ???? ?????????', hi
'KheISutra.guru ?? ????? ???', da 'Lavet med KheISutra.guru') ? a public, shared, brand-bearing
artifact. These 5 short strings are NOT covered by #77 (SPA console) or A3 (site) and need the
same native sign-off. Cheapest of the three review items (5 strings).",
    "owner_gated": true
  },
  {
    "title": "Owner font-glyph eyeball on a real Kannada/Hindi device (both surfaces + a
downloaded reel)",
    "priority": "P1",
    "effort": "S",
    "why": "FRIENDLY $9 DoD gates + MULTILINGUAL EX2 ? cloud CI cannot render font glyphs,
so the owner must confirm Noto Kannada/Devanagari display correctly with no tofu on SPA + site +
the watermarked reel. Closest existing proxy is the Playwright screencasts in khelsutra-
screencast/, but the doc is explicit this is an owner-machine gate. Pure owner action, no
engineering.",
    "owner_gated": true
  },
  {
    "title": "MULTILINGUAL Phase B (Thai) + Phase C (CJK zh-Hans/ja/ko) ? owner go/no-go on
$8 Q1-Q6",
    "priority": "gated",
    "effort": "L",
    "why": "MULTILINGUAL $7 EB1/EB2/EC0-EC3 are all Todo and blocked at the E0 owner-
approval gate. Six open owner questions ($8): approve A?B?C phasing; zh-Hans-only vs +Hant; CJK
offline-bundle strategy (ship-all-chunks vs subset-build vs online-lazy ? a real tens-of-MB
tradeoff against the air-gapped guarantee); native-reviewer sourcing/budget for 7 langs; market
priority order; whether site/ localizes too. Engineering is well-scoped (foundation proven
locale-pluggable) but cannot start without these answers. NOT a launch blocker for the India
beachhead.",
    "owner_gated": true
  },
  {
    "title": "Decide B7 ? engine backend customer-facing localization (report customer_* +
rejection reasons)",
    "priority": "gated",
    "effort": "L",
    "why": "The single remaining gated row in FRIENDLY $7 and the only thing besides the two
human gates standing between the doc and a DONE/archive flip. Owner-gated (D-BACKEND-DEPTH) +
cross-repo (engine Python, no existing backend-i18n infra). Currently SPA-only A5 maps engine
errors client-side; true localized dynamic engine text is phase-two. Surface for an explicit
defer-or-do decision so FRIENDLY can be archived with B7 as a documented deferral.",
    "owner_gated": true
  },
  {
    "title": "Appliance T5/T6 SPA runnable-tool functionality (progress polling + restart-
failure UX; embed RallyPicker + compile + download)",
    "priority": "P2",
    "effort": "M",
    "why": "LOCAL_APPLIANCE $9 T5 + T6 are the remaining non-gated SPA functional rows; T4
(in-UI upload) is in-progress with the segmenter picker still pending. These are runnable-tool
polish, not l10n ? lower priority than the launch gates above, and the core CUJ already works
live e2e (T12 done, DRY_RUN_CUJ $9). Pick up only after the human review gates are unblocked.",
    "owner_gated": false
  }
],
"stale_doc_corrections": [
  "FRIENDLY $7 milestone-prose (line ~147) says 'A2 WhatsApp/social share is deferred out of
v1 scope ? not a DoD blocker', but A2 actually SHIPPED as #82 (verified in git log + the doc's
own changelog at 2026-06-22). The $7 prose contradicts the row (A2 = Done) and the changelog;
the deferral note is stale.",
  "FRIENDLY $7 + $9 describe the remaining native review as covering 'es/da/id + the reel-

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watermark strings' as if the watermark surface were still pending wiring. Git confirms the watermark chain is fully LIVE on engine origin/master (LW1 #304 + LW3 #306) + khelsutra (LW2 #73) ? so the watermark phrases are a shipped, user-visible artifact that needs review, not a future surface. The review scope is correct but any 'not yet live' framing is stale.",

"LOCAL_APPLIANCE §9 T2 row still shows Todo ('Appliance shell ... edge auth; rotate key'), but per session context T2 (alpha deploy) is effectively DONE ? the split-topology alpha is deployed + live-verified (kit merged khelsutra #78/#79/#81/#85). The §9 row predates the split-topology pivot; the truth is captured in the §9 'Alpha deploy path chosen = SPLIT' note below the table but the T2 row checkbox is stale.",

"FRIENDLY §9 DoD note (line ~165) referenced the own-model English-on-kn/hi gap as an open 'engineering follow-up (3)'; it is now CLOSED by #86 (own-model body fully localized) ? the doc's own final changelog entry confirms this, so any earlier 'remaining slice' phrasing is superseded."

```
]
},
{
  "area": "engine serving / platform / owned-model",
  "current_status": "COMPUTE_DECOUPLED_SERVING (CDS, ADR-014) is the active platform track and is the successor to the old \"async job model\" slice. Verified against engine origin/master (head a42c132, log -60) and docs/COMPUTE_DECOUPLED_SERVING/README.md §7 (source-of-truth tracker):\n\nSHIPPED (default-OFF, code on master, all in CDS §7): M0 design APPROVED 2026-06-21 (#278); M1 profile/compute_target config (#279); M2 URI-aware input via StorageRef (#280); M3 configurable output/scratch base, default stays output/ (#282); M4 DeviceContext + native-WASB CPU fallback (#283); M5 truly-local preset + per-profile startup checks + L4 parity test (#286, runlog still owner-GPU-gated); M6 Cloud Run JOB dispatch shim + worker image + mocked tests (#287, backend/compute/cloud_run.py); M8 per-job cost ledger + pricebook + /cost API (#288, DB v6); M9 Cf-Access edge-auth + per-tenant owner_id (#290) PLUS the exposure-safety boot posture in backend/config/startup.py (#294, the current working-tree branch) ? edge-auth-ON refuses to boot without the allowed_video_root jail + CF team/AUD. Also shipped: GET /api/health + download_url from /api/compile (#268); T11 retention sweep (#299); DB now at v7 (locale-per-job, #306).\n\nA REAL L4 cloud GPU run happened 2026-06-21 (owner-authorized, within the $100/D17 cap): g2-standard-4 in asia-southeast1, native WASB ? 22 rallies ? gs:// outputs, VM deleted (~$0.21). ? the no-Gemini fully-local cloud path needs indexing.skip_ai_handoff=true (the long-standing \"0 rallies\" cold-start fix). BUT this was a manual SSH/CLI run, NOT the M7 Cloud-Run-SERVICE deploy ? M7 (persistent reachable endpoint + DEPLOY_REPORT) is still ? todo.\n\nNOT shipped (verified absent on master): M7 (cloud-serving profile e2e / persistent service); M11 flywheel (no backend/flywheel.py); M12 gdrive-api OAuth StorageBackend (only the mount-only backend/storage/gdrive.py exists, no gdrive-api:// scheme).\n\nUI-driven-cloud-serving gap: the SPA ingest/upload?/api/process?poll /api/jobs?reel flow IS BUILT in khelsutra (web/src/components/IngestPanel.tsx + hooks/useIngestJob.ts + useUpload.ts, with tests; B-P2 #56 + T4 #66). The genuinely-missing UI piece is the compute PICKER (\"your machine vs our cloud\", D13/D15) ? no compute_target selector exists in the SPA. The alpha is live on the CPU droplet via the Gemini path (per session context), so the cloud-GPU half of the picker has no reachable persistent endpoint behind it yet (that is M7).\n\nOWNED-MODEL track: docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md is still PLAN/roadmap ? NO implementation has started; Phase-0 is per-milestone owner-greenlit ($0/local). Recommended first milestones in the doc: M0.1 (labeled-corpus spine + event-index contract) then M0.4 (Gen-0 TAL harness; the doc says ActionFormer but the 2026-06-17 external review + NEXT_STEPS prefer TemporalMaxer). The whole track is blocked on growing the golden corpus (owner).",
  "next_items": [
```

```
{
  "title": "M7 ? stand up the cloud-serving profile as a persistent Cloud Run service + write the DEPLOY_REPORT",
  "priority": "P1",
  "effort": "M",
  "why": "M7 is the ? keystone the rest of CDS depends on (M8 real prices, M9 consent, M11 flywheel, M12 Drive all list M7 as a dependency). M6 dispatch code + worker image already merged (#287) and a real L4 run already proved the worker (22 rallies, $0.21). What's left is the real build/push + a reachable endpoint + the DEPLOY_REPORT runlog. Owner-gated on GCP spend but inside the already-authorized $100/D17 cap.",
  "owner_gated": true
},
{
  "title": "Add the compute PICKER to the khelsutra SPA (local vs cloud, D13/D15)",
  "priority": "P1",
```

```

    "effort": "M",
    "why": "This is the ONLY genuinely-unbuilt piece of the owner-priority 'make it UI-
driven' gap ? the ingest/upload/poll/reel SPA flow already shipped (B-P2 #56, T4 #66), contrary
to the stale NEXT_STEPS P0 item. The picker surfaces compute_target (M1 config already exists)
so a user confirms 'your machine vs our cloud' at start (D15: auto-detect suggests, user
confirms). Cross-repo (khelsutra web/); the cloud option is only useful once M7's endpoint
exists, so sequence after/with M7.",
    "owner_gated": false
  },
  {
    "title": "M11 ? data-flywheel harness (capture?store?shadow-eval?goldenize), behind
flywheel.*=OFF",
    "priority": "P2",
    "effort": "L",
    "why": "Closes the loop that lets the owned model climb to the D11 ship-gate floor. NOT
yet shipped (no backend/flywheel.py on master). Reuses workspace.for_video +
backend/eval/experiment.compare + promotion.py. Privacy-sensitive: needs a real consent gate
(M9-consent) and the consent columns are a NEW v8 migration ? NOT v7 as NEXT_STEPS still claims
(v7 was taken by the locale column in #306). Depends on M7 + M9-consent; ToS/DPA text is
owner/counsel.",
    "owner_gated": true
  },
  {
    "title": "M12 ? gdrive-api OAuth StorageBackend (Drive round-trip Northstar D14)",
    "priority": "P2",
    "effort": "L",
    "why": "Net-new and verified absent (only the mount-only gdrive.py exists; no gdrive-
api:// scheme). Realizes the D14 mobile/cloud-native Northstar: read source from + write reel
back to the user's Drive. Slots into the StorageBackend ABC (mirror gcs.py DI against an
injected fake Drive). Owner-gated on the Google OAuth app registration + per-user consent;
depends on M7 + M9.",
    "owner_gated": true
  },
  {
    "title": "M8/M9 owner residuals ? calibrate the pricebook with real GCP prices+FX, and
supply ToS/DPA + CF team-domain/AUD",
    "priority": "P2",
    "effort": "S",
    "why": "M8 ledger/pricebook (#288) and M9 edge-auth (#290) shipped default-OFF but carry
owner-side residuals before they can run real: M8 needs real GCP prices + FX + usage-emission;
M9 needs counsel ToS/DPA consent text + the CF team-domain/AUD + ingress lock. These don't block
Claude's default-OFF code but block turning the features ON for real tenants.",
    "owner_gated": true
  },
  {
    "title": "Owned-model Phase-0: greenlight M0.1 (corpus+event-index) then M0.4 (Gen-0 TAL
harness, prefer TemporalMaxer)",
    "priority": "gated",
    "effort": "L",
    "why": "OWNED_MODEL_IMPLEMENTATION_PLAN.md is still PLAN-only ? no implementation has
started and nothing starts without an explicit per-milestone owner greenlight. The whole track
is bottlenecked on the owner growing the golden corpus (multi-court badminton + TT). Recommended
first milestone = M0.1; the M0.4 harness should use TemporalMaxer over ActionFormer per the
2026-06-17 external review. $0/local once greenlit.",
    "owner_gated": true
  }
],
"stale_doc_corrections": [
  "DEFERRED_HARDENING_PLAN.md #16 'async job model' is NOT the committed-next slice as the
task framing assumed ? it is DONE (the whole doc is a COMPLETED pointer stub: #16 shipped in PR
#87, hardened by the #149 drain loop). The committed-next platform slice has moved on to the
COMPUTE_DECOUPLED_SERVING subproject.",
  "NEXT_STEPS.md P0 item #0(ii) lists 'the SPA ingest/progress screen (khelsutra web B-P2:
enter/upload video ? /api/process ? poll /api/jobs ? reel)' as a pending need. STALE ? it
already shipped: bucket ingest B-P2 (khelsutra #56) + in-UI file upload T4 (#66), with

```

IngestPanel.tsx + useIngestJob/useUpload hooks and tests. The only unbuilt UI piece of that gap is the compute picker (D13/D15).",

"NEXT_STEPS.md P1 item #4 says 'M9-consent cols = a new v7 migration (M8 took v6)'. STALE ? v7 has since been taken by the locale-per-job column (#306, on master). The M9-consent / M11 columns now need a NEW v8 migration.",

"NEXT_STEPS.md P1 item #4 lists 'M5/M6/M8/M9-auth ? merged' but still lists 'REMAINING: M11 + M12' under a single batch; verified accurate that M11 (backend/flywheel.py) and M12 (gdrive-api scheme) are genuinely absent on master ? flagging only to confirm they were correctly NOT counted as shipped.",

"NEXT_STEPS.md 'committed next PLATFORM slice = async job model (single-tenant)' (final bullet) is stale relative to the 2026-06-21 header it carries ? async jobs shipped; the live committed platform work is CDS M5?M12. The OWNED_MODEL plan's identical closing line is also outdated in the same way."

```
]
},
{
  "area": "rally-detection quality",
  "current_status": "SHIPPED on engine origin/master (verified via git log + code, NOT just docs): the held-shuttle/Gate-A question and the multi-signal-fusion scaffolding are DONE, and the windowing R-series has already LAUNCHED a default-ON milestone ? so the detailed NEXT_STEPS \"next\" list is stale. Specifics:\n- Gate A is built (backend/pipeline/detectors/rally_gate.py) and the live FusionSegmenter (backend/pipeline/segmenters/fusion_hybrid.py, registered `fusion_hybrid`, default-OFF) can apply it via config knobs (config/models.py: min_crossings/min_players default 0=OFF). The owner's \"held-shuttle counted as rally\" concern is NOT a missing capability.\n- The Gate-A served-default FLIP (#146) was tried, REGRESSED in production (eval?serve), and was REVERTED (#151), then reconciled in docs (#152/#153) and the eval?serve gap closed with a single-sourced SERVED preset + promotion guard (#154). So Gate A is correctly kept OPT-IN, not default ? that loop is CLOSED, do not relitigate.\n- R2 task-faithful tolerance metric (tolF1, asymmetric ?_start=2/?_end=1.5) SHIPPED (#199, design #198).\n- R4 rally-END inactivity trim SHIPPED (#200).\n- R1 MILESTONE LAUNCHED and FLIPPED DEFAULT-ON (#204): config/models.py now defaults max_window_duration=6.0 (W1 cap=6) + R4 (window_inactivity_end=1.5). Re-verified at n=13 golden: baseline?milestone ?tolF1 +0.222, sign 12/0/1, p=0.0005; cap6 beats cap12 +0.110; +R4 +0.040 (p=0.022). Net over-seg promotion guard (#201) + nightly CPU regression tripwire (#202) + R1 Launch Report all landed. This is the biggest quality win and it is ALREADY IN PRODUCTION DEFAULTS.\n- Person-cue fusion P1 DONE (PersonDetector); P3 person-feature harness machinery exists (fusion_compare.py/fusion_golden.py/ablation.py) but the real LOVO measurement is owner/GPU-gated (needs GPU-extracted golden features + .rallies.csv). Audio cue (P4) still scaffolded/deferred.\n- Still default-OFF and either UNMEASURED or measured-not-promoted: W2 net-crossings gate (#188), hard-cap fallback (#195), R5 blob-fallback (#203, helps only the 2 continuously-active blob clips, corpus-mean n.s.), P7 long-rally lens + stitching.min_rally_duration knob (#220), and the prominent-court gate (#243).\nFrontier per QUALITY_ITERATIONS.md (newest entries P7 + prominent-court): with the R1 milestone live, single-court over-fire is largely tamed (seg_ratio ~1.04 on long>5s), but MULTICOURT still over-fires 1.58x ? the P7 \"FPs are short\" thesis FAILS on multicourt; adjacent/background-court rallies are the #1 confirmed remaining gap, and they need a COURT signal, not a bigger cap.",
  "next_items": [
    {
      "title": "Measure + promote the prominent-court gate (#243) on the golden corpus ? the cheapest OPEN quality win",
      "priority": "P1",
      "effort": "M",
      "why": "court_gate.py (#243) is already BUILT and wired into rally_seg_eval.py (default absent court_polygon = no-op, so zero risk shipped) but has NO measured golden entry in QUALITY_ITERATIONS.md ? it is the only built-but-unmeasured lever aimed squarely at the #1 confirmed remaining gap: multicourt background-court FPs (P7 thesis FAILS, 1.58x over-fire). Needs per-clip court polygons for the multicourt golden clips + an A/B through the harness; promote default-OFF?ON only on measured tolF1 lift via run_regression with the over-seg guard + a Launch Report. Pure-eval + harness logic is testable locally; final golden numbers need owner-side GPU features.",
      "owner_gated": true
    },
    {
      "title": "Run the real P3 person-feature LOVO measurement on the 6+ golden videos (fusion_compare/ablation)",

```

```

    "priority": "P1",
    "effort": "M",
    "why": "The P3 machinery (fusion_compare.py, fusion_golden.py, ablation.py with
bootstrap CI + paired sign test) is shipped and suite-green, but the actual person-cue ?F1 is
unmeasured ? blocked only on GPU-extracted *_golden_features.json + .rallies.csv labels
(owner/GPU box). Person/court localization is the architecturally-correct fix for the multicourt
gap (Gate B + court mask). This is the gating data step before any person-fusion promotion.",
    "owner_gated": true
},
{
    "title": "Grow the golden corpus ? multicourt badminton first, then >=3 matches/sport
(TT)",
    "priority": "P0",
    "effort": "L",
    "why": "The single highest-leverage move and the explicit gate on nearly every quality
lever. The milestone re-verified at n=13/n=15; the paired sign test reaches p<0.05 with more
multicourt folds, and the multicourt ship-target (wrapper 0.66) can only be chased with more
multicourt data. Owner-only (golden labels + videos are owner-side). Pure owner action, not
Claude-doable.",
    "owner_gated": true
},
{
    "title": "Owner: ratify the ship-gate target (#172) now that R2 tolF1 is the metric",
    "priority": "P1",
    "effort": "S",
    "why": "R2 tolF1 shipped as the task-faithful metric and the R1 milestone gave a
concrete baseline (multicourt reference = Gemini wrapper 0.66, floor = heuristic). A ratified
gate (e.g. 'LOVO tolF1>=0.70 on multicourt folds, paired-sign p<0.05 vs heuristic') turns
run_regression's promotion guard from advisory into a real release gate. Owner decision; needs
corpus growth to be meaningful.",
    "owner_gated": true
},
{
    "title": "Evaluate TemporalMaxer as the M0.4 learned-segmenter swap (parameter-free,
small-N robust)",
    "priority": "P2",
    "effort": "M",
    "why": "Per the 2026-06-17 external due-diligence review (RALLY_QUALITY_RESEARCH.md
$5.6), prefer TemporalMaxer over ActionFormer/ASFormer (~2.8x fewer GMACs, parameter-free max-
pool) for the cheap/local/fast mandate ? a $0/local increment over the Gen-0 LogReg inside the
existing harness. Deferred until the cheaper windowing+court+person levers are exhausted and
corpus is larger; do not start without owner go.",
    "owner_gated": true
}
],
"stale_doc_corrections": [
    "docs/NEXT_STEPS.md detailed quality-track section (dated 2026-06-13/06-15) still lists
'Wire Gate A into production served path' and 'P3 first step (this work item): harness compare
LOVO for person-on variant' as the pick-up ? both are SUPERSEDED: Gate A wiring +
FusionSegmenter shipped (#146 flip tried, reverted #151, kept opt-in #152-#154), and the P3
harness machinery shipped (fusion_compare/ablation present; only the GPU MEASUREMENT remains).
The doc's own top header (2026-06-21) flags the lower section as historical, but the detailed
pick-up bullets read as still-next and would mislead.",
    "The stale section frames 'Wiring Gate A into the live segmenter ... was the single
cheapest, highest-confidence quality win' as the pending top win. That win was REALIZED via a
different lever: the R1 windowing milestone (cap=6 + R4) is what actually shipped default-ON
(#204, n=13, p=0.0005) and is the dominant quality lever ? Gate A on production windowing was
measured neutral/negative and kept opt-in. The current cheapest OPEN win is measuring+promoting
the prominent-court gate (#243), not wiring Gate A.",
    "docs/HOW_RALLY_DETECTION_WORKS.md 'three knobs' (velocity_thresh / merge_gap /
min_window_duration) understates the live segmentation layer: as of #187/#200/#204 the
production defaults also include bounded windowing (max_window_duration=6) + R4 rally-end
inactivity trim, plus opt-in Gate A/B + R5 blob-fallback + prominent-court gate knobs. The doc
is conceptually fine but no longer enumerates the shipped knob surface.",
    "Several QUALITY_ITERATIONS.md / NEXT_STEPS references to corpus size are inconsistent

```


across entries (n=6, n=11, n=13, n=15) ? the latest authoritative numbers are the R1 launch at n=13 (#204) and the P7 stratified lens at n=15 (#220). Use n=15 as the current golden size when scoping."

```
]
},
{
  "area": "This-session follow-ups + cross-cutting hygiene",
  "current_status": "See next_items and stale_doc_corrections for grounded detail.",
  "next_items": [
    {
      "title": "Finding B: fail the job on missing GEMINI_API_KEY or unrunnable config instead of a silent 0-rally success",
      "priority": "P0",
      "effort": "M",
      "why": "gemini.py:92-99 returns None on a missing key, then process_video returns an empty list at :514-516; job_worker.py:76-90 marks status=done. On the alpha box the tester gets a silent empty result and the operator only sees the log. Distinguish a genuine no-rallies result from an unrunnable config (missing key, unknown provider, unknown sport) and either raise so the worker marks status=failed with a reason, or set a result error the SPA shows. Mirrors gemini_refine.py SystemExit (CODE_MAP:179), extends the #271 0-segment precedent, matches the loud-failures launch bar. One-PR hardening item; serving owns backend/config and the gemini segmenter.",
      "owner_gated": false
    },
    {
      "title": "Auto-schedule the retention sweep on the live droplet (engine #301)",
      "priority": "P1",
      "effort": "S",
      "why": "#299 shipped tools/cleanup_caches.py (dry-run, uploads over 7d, --json cron-ready) but #301 (scheduling) is OPEN. The alpha box ingests real uploads, so footage lingers until --apply is run by hand vs the 7d privacy commitment. A systemd timer or cron running --apply in the droplet kit (#81/#85) closes it.",
      "owner_gated": false
    },
    {
      "title": "Revoke or scope-down the broad cloudflared login cert.pem left on the alpha box",
      "priority": "P1",
      "effort": "S",
      "why": "DEPLOY_ALPHA s5 cloudflared tunnel login writes an account-wide cert.pem (zone-edit and tunnel-create) on the box; only the per-tunnel UUID.json is needed to run the tunnel. s8 teardown never revokes it. Owner: delete cert.pem (or rotate the origin cert) after create; add a note to s5/s8. Matches LOCAL_APPLIANCE_ARCHITECTURE:137 credential-leak guidance.",
      "owner_gated": true
    },
    {
      "title": "Native-speaker review of es/da/id + reel-watermark strings + owner kn/hi font eyeball (wide-launch gate)",
      "priority": "P1",
      "effort": "M",
      "why": "FRIENDLY_AND_MULTILINGUAL s9 line 147 is the last open DoD item before WIDE launch; the kn/hi SPA console review shipped (#77) but es/da/id and watermark strings are AI-drafted and the font render check (lines 159/161) is an owner-eyeball gate. Alpha allowlisted testers do not block on it; it gates dropping the allowlist. Owner-coordinated.",
      "owner_gated": true
    },
    {
      "title": "Basic live-droplet monitoring (uptime + job-failure + Gemini-cost alerts)",
      "priority": "P2",
      "effort": "M",
      "why": "Single CPU droplet, no documented monitoring; obsreport (BACKLOG:85-89) is per-run, not service health. The SPA shows a client-side offline banner (DEPLOY_ALPHA s7) but nothing alerts the OWNER on a box or tunnel drop, job failures, or Gemini spend spikes. The Ops-Agent (#291) is GPU-VM-only, not this droplet. Light touch: /api/health ping behind Access, a log-scan or the Finding-B failed-job signal, a Gemini-cost ceiling. Pairs with Finding B.",
    }
  ]
}
```

```

    "owner_gated": false
  }
],
"stale_doc_corrections": [
  "Finding B is NOT covered by exposure-safety preflight #294 (HEAD 6fdd1d4): it only checks
exposed-server boot posture (allowed_video_root jail, cf_team_domain/cf_access_aud, PyJWT) and
is default-OFF; it never touches Gemini key absence or the 0-rally outcome. Genuinely uncovered,
not already-fixed.",
  "Finding B is absent from BACKLOG.md (stamped 2026-06-10, pre-deploy), DOC_STATUS.md, and
NEXT_STEPS.md; the closest is the shipped per-video obsreport (a different layer). Should be
tracked as a deferred or hardening item.",
  "Retention reads as done in NEXT_STEPS and khelsutra trackers, but #299 shipped only the
dry-run TOOL; auto-scheduling #301 is OPEN, so it is a half-truth on the live box until a
scheduler runs --apply.",
  "Ops-Agent monitoring (#291 / NEXT_STEPS P1 item 6) is GPU-VM-only; it does NOT cover the
alpha CPU droplet, so the live-box monitoring gap is real despite the work appearing done.",
  "#301 is an ENGINE issue (auto-schedule the retention sweep) referenced from khelsutra
docs with no engine commit on origin/master; still open work, not merged."
]
}
]

```

2. assistant (2026-06-22T02:04:05.706Z)

I'll verify the key claims against the actual repo state before synthesizing. Let me start with the high-leverage checks: the packaging entrypoints, the "P3 done" memory-vs-master discrepancy, and the Finding B silent-success path.

3. user (2026-06-22T02:04:07.344Z)

```

a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers
(#309)
de2bf14 feat(pkg): P2e ? lazy-torch so the LIGHT tier imports with NO torch installed (default-
OFF) (#308)
679e642 feat(pkg): P2d ? DB upgrade-safety (downgrade guard + backup-before-migrate + export)
(default-OFF) (#307)
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
(#306)
fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place)
(default-OFF) (#305)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
ca34787 feat(pkg): P2b ? friendly `indexer --app` launcher (boot localhost server + open the
bundled UI) (#303)
4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-
OFF) (#302)

```

4. user (2026-06-22T02:04:08.746Z)

```

24b5dee docs(serving): mark $7 tracker M9 ? + changelog (PR #290) + the real M7 L4 run (22
rallies)
fefa412 docs(serving): resolve $8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
31213e9 docs(serving): resolve $8 design picks (D16/D17) + reconcile per-video P3 vs M3
b649f1c address review on #169: revert frozen archive edit (should-fix); trim handoff to short
pointer per #165 intent (record moved to plan); add Transition Record section; unbundle catalog
items 3/6/10 + clarify P3 'shipped' state with investigation (no fusion_compare.py etc. present;
general harness + P2 are); add M0 Quick Wins (the 4 in-container items per course-correction);
re-sequence milestones (M0 quick wins first + owner-led parallel, heavy last); update risk table
+ sequencing text. Thorough point-by-point in PR comment.
c558372 Merge pull request #166 from avidullu/test/rally-quality-p3-harness
5fabfffd address review for #166: retitled PR (via update); restored Gate-A cheapest-win
rationale, P2 FusionSegmenter specifics, served regression test finding/revert note in
NEXT_STEPS; removed long P3 prose comment from fusion_hybrid.py source (moved essence to docs);
confirmed archived snapshot Post-merge #165 addition is one-time finalization (no further edits
to frozen archive). Fixes per owner review points. Ready for re-review.

```

5cc0ea4 test + docs: hardening loop completion record + rally quality track P3 first step (harness LOVO for person features); identify next candidates (per NEXT_STEPS). Small landing comment in fusion_hybrid + explicit list in NEXT_STEPS. Syntax green. Pre: container baseline (ast + obsreport note). Branched, one item, will babysit. Verification: owner re-run full + real video.

918a4ff feat(jobs): worker self-scheduling drain loop honoring WAIT_AND_RESUME (EXECUTION_POLICY P3a) (#149)

5cb7bbf feat(jobs): worker self-scheduling drain loop honoring WAIT_AND_RESUME (EXECUTION_POLICY P3a)

ec5bbe0 chore(eval): track experiment_leaderboard.json (fusion P3/P4 honest LOVO records) (#139)

74c1303 docs(quality): log fusion P3+P4 measured ?F1 on the n=6 golden corpus (#137)

39e961b feat(eval): fusion P3 measurement ? golden person-feature extractor + LOVO compare (#121)

5. assistant (2026-06-22T02:04:13.198Z)

Memory says "P3 GPU validation DONE" but there's no such commit on master. The scan's stale-doc correction is confirmed directionally. Let me verify the packaging entrypoints, the Finding B silent-success path, and the PACKAGING_PLAN "? NEXT" block directly.

6. user (2026-06-22T02:04:16.164Z)

```
1:# PEP 621 packaging for the badminton/sports highlight indexer (DEFERRED #7).
17:name = "badminton-highlight-indexer"
81:# `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
82:indexer = "backend.main:main"
---APP FLAG---
```

7. user (2026-06-22T02:04:17.344Z)

```
ls: cannot access 'backend/ui/': No such file or directory
---SPA mount---
30:from fastapi.staticfiles import StaticFiles
87:app.mount("/static", StaticFiles(directory=_STATIC_DIR), name="static")
```

8. assistant (2026-06-22T02:04:26.507Z)

Important discrepancy. The local working tree is on branch `serving/exposure-safety-preflight` (per git status), which is BEHIND the packaging work. The scan claims are about `origin/master`. Let me check the actual `origin/master` state, since the local checkout doesn't have `backend/ui/` or the `--app` flag.

9. user (2026-06-22T02:04:27.731Z)

```
serving/exposure-safety-preflight
---HEAD---
6fdd1d4 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed engine (default-OFF)
---origin/master HEAD---
a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers (#309)
```

10. user (2026-06-22T02:04:30.031Z)

```
=== pyproject entrypoint on origin/master ===
86:indexer = "backend.main:main"
=== --app flag on origin/master ===
131:# --- P2: the friendly `indexer --app` launcher (cross-platform, no native component ? 01)
---
152:     """For --app: keep `port` if free, else fall back to a free one so a busy 8000 never
blocks the app."""
158:     Returns True if a browser open was attempted. No-op (returns False) when
RALLY_NO_BROWSER is set
160:     if os.environ.get("RALLY_NO_BROWSER"):
```

```

175:         webbrowser.open(url)
867:     parser.add_argument("--app", action="store_true",
981:         # CLI processing mode (no web UI needed). --app/--server always run the server, never
        CLI single-shot.
1074:         # with --app, or the poll 400s and the browser just won't auto-open ? the
        server still runs).
1076:         f"(set RALLY_NO_BROWSER=1 to skip). Press Ctrl+C to stop.")
=== backend/ui on origin/master ===
100644 blob 6c4ef042e0f7f1b0aec0c4eb3e8129856f467189    backend/ui/THIRD_PARTY_NOTICES.md
040000 tree bfb230dbe062f537128f5a30d6e53cbb62b70078    backend/ui/assets
100644 blob 61378a9640e76b8f598ed1dfc45ef874e4f01c51    backend/ui/index.html
100644 blob 51a9859f6f0e8e653c7bfcd0ca76ab070f17fc97    backend/ui/ui_version.json

```

11. assistant (2026-06-22T02:04:33.530Z)

Confirmed: `origin/master` has the `--app` flag, bundled SPA (`backend/ui/`), and the `indexer` entrypoint ? all as the scan claims. The local working tree is just on an older serving branch. Now let me verify the Finding B silent-success path and the packaging "remaining gap" (P2b-install) on origin/master.

12. user (2026-06-22T02:04:36.221Z)

```

=== gemini.py missing-key path ===
83:         return None
90:         api_key_env_var = f"{provider_name.upper()}_API_KEY"
91:         api_key = os.environ.get(api_key_env_var)
92:         if not api_key:
93:             from backend.pipeline.segmenters._keyhint import api_key_hint
95:             f"{api_key_env_var} environment variable is not set! "
97:             + api_key_hint(api_key_env_var)
99:         return None
102:         provider = provider_registry.get(provider_name, api_key=api_key,
        model_name=model_name)
105:         return None
516:         return []
554:         return []
=== job_worker status=done path ===

```

13. user (2026-06-22T02:04:39.097Z)

```

=== uv tool install / install script / wizard on origin/master ===
origin/master:backend/main.py
origin/master:backend/utils/ffmpeg.py
origin/master:tests/test_api_endpoints.py
=== /api/capabilities ===
339:@app.get("/api/capabilities")
340:def capabilities():
=== PACKAGING_PLAN NEXT block ===
11:## ? NEXT SESSION ? start here
15:1. **P3 GPU validation + screencast (owner wants first; ~$0.50, well under the $100 cap).**
Provision an L4 (`deploy/gcp/create_gpu_vm.sh`) ? install the **wheel** on the fresh GPU box +
the WASB env ? validate the packaged stack end-to-end (`indexer --app` ? bundled SPA ? native-
WASB ? reel) ? **screencast** ? `teardown.sh` (deletes) + confirm $0. ? Run as a FOCUSED op
(billable VM ? teardown is priority #1). ? Known blocker: native-WASB returned **0 rallies** on
a fresh build last session ? reproducing the 22-rally result needs the engine session's exact
tuned `worker infer detector_impl=native` config. Confirm cmd + cost before spend; `[[gcp-vm-
always-delete]]`.
16:2. **P2b-install** ($7) ? `uv tool install` / a generated cross-platform install script (now
a clean **torch-free** LIGHT install thanks to P2e) + ship a truly-local default `config.json` +
an uninstall manifest.
17:3. **P2c first-run wizard** ($7) ? the 3-step wizard UI (Gemini key / compute / videos) in
the **khelsutra `web/` SPA**, rendered from `GET /api/capabilities`. **Cross-repo**
(`avidullu/khelsutra`) ? coordinate like P1b (the SPA is gitignored-dist; refresh `backend/ui/`
via `scripts/bundle_ui.sh` after).

```

```
113:| P2b-install | **Install path** (`uv tool install` / a generated cross-platform install
script; ship `config.json` defaulting truly-local; optional thin per-OS shortcut to `indexer
--app`) | P2b-launcher | No (signing only if clickable binaries ? 09) | ? | ? |
114:| P2c | **In-UI 3-step first-run wizard** (Gemini key / Compute / Videos), rendered from
`/api/capabilities`, en/kn/hi | P1b | No | ? | ? |
115:| P2d | **DB upgrade contract** ? `SCHEMA_VERSION` + downgrade guard (`SchemaTooNewError`) +
backup-before-migrate (`<db>.bak-v<old>`, WAL-complete) + `backup_database()` export. Byte-
identical for fresh/current DBs. *(uninstall manifest deferred to P2b-install.)* | ? | No | ? |
[#307](https://github.com/Khelsutra/badminton-highlight-indexer/pull/307) |
```

14. assistant (2026-06-22T02:04:47.310Z)

All key claims confirmed against `origin/master`:

- Packaging: `--app` flag, `backend/ui/` bundled SPA, `indexer` entrypoint, `/api/capabilities` all present; P2b-install and P2c wizard genuinely absent (no `uv tool install`/install script/wizard outside docs/tests). PACKAGING_PLAN's ? NEXT block lists P3 GPU validation as item #1 with the "0 rallies" blocker ? contradicting the MEMORY.md "P3 DONE" claim.
- Finding B: `gemini.py:99` returns None on missing key ? `:516` returns `[]`. Confirmed the silent-empty path exists.

Let me confirm the job_worker marks status=done on empty results, and verify the R1 quality milestone default-ON flip (the claimed biggest quality win).

15. user (2026-06-22T02:04:48.941Z)

=== job_worker outcome on empty ===

16. user (2026-06-22T02:04:51.109Z)

=== R1 windowing defaults on origin/master ===

```
100:   threshold is raised (`min_crossings`/`min_players` default 0 = OFF). The court mask
101:   is optional: `court_polygon=None` ? Gate B counts every detection (player-count-only);
115:   min_crossings: int = Field(default=0, ge=0)
118:   min_players: int = Field(default=0, ge=0)
120:   court_polygon: Optional[list[list[float]]] = None
155:   max_window_duration: float = 6.0
159:   # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
169:   window_inactivity_end: float = 1.5
174:   # than window_blob_factor × max_window_duration (no internal gap, no rally-end signal ?
e.g.
181:   # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
=== court_gate present? ===
backend/compute/cloud_run.py
backend/pipeline/detectors/court_gate.py
backend/pipeline/detectors/rally_gate.py
backend/pipeline/segmenters/fusion_hybrid.py
backend/storage/gdrive.py
tests/test_court_gate.py
tests/test_fusion_hybrid.py
tests/test_rally_gate.py
```

17. assistant (2026-06-22T02:04:54.772Z)

Confirmed: R1 windowing defaults are default-ON (`max\_window\_duration=6.0`, `window\_inactivity\_end=1.5`); Gate A/court knobs default-OFF (0); `court\_gate.py`, `cloud\_run.py` exist; `flywheel.py` absent (M11 not shipped). Let me find the job_worker file and confirm the done-on-empty path, plus confirm M7/M11/M12 absence.

18. user (2026-06-22T02:04:57.462Z)

=== job worker files ===
backend/api/job_worker.py

```
tests/test_job_worker_loop.py
tests/test_job_worker_module_seam.py
=== flywheel absent? ===
(empty above = absent)
=== gdrive-api scheme? ===
5:needs no service account, no API quota, no OAuth, and Drive streams + caches bytes on access.
So
```

19. user (2026-06-22T02:05:01.176Z)

```
=== job_worker.py outcome handling ===
5:that raises `JobWaiting` is parked (status 'waiting' + `next_poll_at` + progress) and
73:  """Run ONE job already claimed (`status='running'`) by `claim_due_job`, recording its
76:  Job `status` is the EXECUTION state (queued|waiting|running|done|failed): 'done' means
78:  result["status"]; 'failed' means the worker itself crashed/raised; 'waiting' means the
86:      result = _main._execute_processing(
88:      if result.get("video_id"):
89:          db_instance.set_job_video_id(job_id, result["video_id"])
90:          db_instance.mark_job_done(job_id, result)
91:          _maybe_record_job_cost(job, result) # M8 cost ledger (default-OFF)
92:  except _main.JobWaiting as w:
96:  except Exception as e:
97:      logger.error(f"Job {job_id} crashed: {e}\n{traceback.format_exc()}")
98:      db_instance.mark_job_failed(job_id, f"{type(e).__name__}: {e}")
101:def _maybe_record_job_cost(job: Dict[str, Any], result: Dict[str, Any]) -> None:
103:  ``config.billing.enabled``. When on, compute the job's cost from ``result['usage']``
107:  ? **Honest gap (M8 scope):** the pipeline does **not** emit ``result['usage']`` yet, so
even with
114:  claimed row (carries ``owner_id``); ``result`` is the pipeline output."""
121:      usage = (result or {}).get("usage") or {}
124:      gpu_type=usage.get("gpu_type"), compute_backend=(result or
{}).get("compute_backend"),
128:  except Exception as e: # never wedge a completed job on bookkeeping
129:      logger.warning("Cost recording for job %s failed (non-fatal): %s", job.get("id"), e)
136:  Each parked job (`JobWaiting` ? status 'waiting') drops out of this drain and is picked
163:  except Exception as e: # pragma: no cover - defensive; never wedge the worker
164:      logger.error(f"Could not compute next-due delay: {e}")
```

20. assistant (2026-06-22T02:05:06.474Z)

Finding B confirmed precisely: `job\_worker.py:90` calls `mark\_job\_done(job\_id, result)` whenever `\_execute\_processing` returns without raising. Since `gemini.py` returns None on a missing key ? empty segment list ? a normal (non-raising) return, a missing `GEMINI\_API\_KEY` yields `status=done` with zero rallies, not `status=failed`. The worker only marks `failed` on an actual exception (line 98). The silent-success path is real and uncovered by the exposure-safety preflight (#294).

All major load-bearing claims in the scans are git-verified against `origin/master`. I have enough to synthesize the answer.

21. user (2026-06-22T02:06:07.378Z)

Structured output provided successfully

22. assistant (2026-06-22T02:06:17.163Z)

I've completed the git-grounded analysis and returned the structured answer. Summary of what I verified directly against `origin/master` (not the stale local serving branch):

- **Packaging works today for a dev** ? `pip install` wheel ? `indexer --app` ? bundled `backend/ui/` SPA in the browser. The `--app` launcher (main.py:867), bundled SPA, and `indexer` endpoint are all confirmed present. **No installer, no wizard** ? P2b-install and P2c are genuinely absent.
- **Finding B is real and load-bearing** ? `gemini.py:99` (None on missing key) ? `:516` (empty

list) ? `job\_worker.py:90` `mark\_job\_done` (only marks failed on an exception at :98). A missing Gemini key silently produces a "done" zero-rally job. This is the top-ranked item because it breaks the very first real-alpha experience.

- **The "P3 DONE" memory claim is unconfirmed on master** ? no such commit; PACKAGING_PLAN still lists it as remaining with a live "0 rallies" blocker. Flagged as the #1 watch-out.
- R1 windowing is default-ON (`max\_window\_duration=6.0`); `court\_gate.py`/`cloud\_run.py` exist; `flywheel.py` (M11) and a `gdrive-api` scheme (M12) are absent ? matching the scans.

The 9 ranked next-steps lead with the two things that unblock real alpha users and the co-founder's download-and-run goal (Finding B, P2b-install, P2c wizard, native review), then sequence the deeper serving/quality/owned-model tracks behind them.